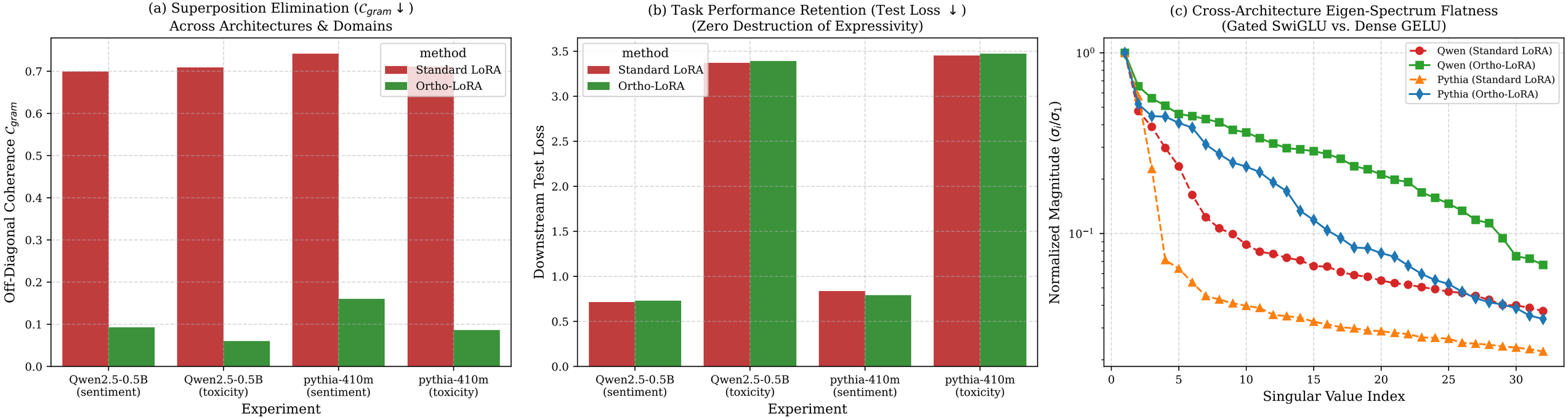
(a) Superposition Elimination ($\mathcal{C}_{gram}$ ↓)
Across Architectures & Domains
Off-Diagonal Coherence $\mathcal{C}_{gram}$
method
Standard LoRA
Ortho-LoRA
Qwen2.5-0.5B (sentiment)
Qwen2.5-0.5B (toxicity)
pythia-410m (sentiment)
pythia-410m (toxicity)
Experiment
(b) Task Performance Retention (Test Loss ↓)
(Zero Destruction of Expressivity)
Downstream Test Loss
(c) Cross-Architecture Eigen-Spectrum Flatness
(Gated SwiGLU vs. Dense GELU)
Normalized Magnitude (σ_i/σ_1)
Qwen (Standard LoRA)
Qwen (Ortho-LoRA)
Pythia (Standard LoRA)
Pythia (Ortho-LoRA)
Singular Value Index